



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

SAFETY DATA SHEET

Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510;

Fax: +66(0)-2721-3515

1. CHEMICAL IDENTIFICATION

Product Identification

Trade name: Petro

Product name: pendimethalin 33% W/V EC

Product use: herbicide

Relevant identified uses of the substance/mixture and uses advised against:

Used as a selective herbicide used to control broadleaf weeds and grassy weed species in a number of crop and non-crop areas and on residential lawns and ornamentals. Pendimethalin is applied by broadcast, chemigation, conservation tillage, containerized plant treatment, soil incorporation, and directed spray.

2. HAZARD IDENTIFICATION

Potential Health Effects

Pendimethalin is colored and will strongly stain skin and clothing. If skin has been exposed to the product, thorough washing with soap and water will remove any excess product. The color on the skin is not deep and will naturally disappear over a few days.

Swallowed: Data suggests that this product should present no significant problems. However, the solvent can be of concern if vomiting has occurred; aspiration of the vomit into the lungs can cause mild to severe pulmonary injury and possibly death.

Eye: This product is mildly irritating to eyes.

Skin: This product is not considered a serious toxic hazard. Prolonged or repeated contact with the concentrate can cause defatting of the skin leading to secondary dermatitis.

Inhaled: High vapor concentration of the solvent, while handling the product are irritating to the eyes and the respiratory tract, may cause headaches and dizziness, can be anesthetic and may have other central nervous system effects. Avoid working in, or inhaling spray mist.



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3. COMPOSITION/INFORMATION ON INGREDIENT

Chemical name: 3,4-dimethyl-2,6-dinitro-N-pentan-3-ylaniline

Common name: pendimethalin

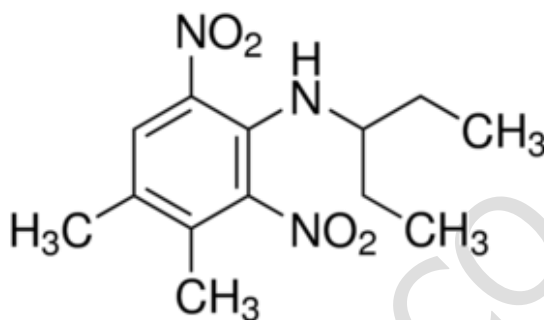
CAS Number: 40487-42-1

Ingredients and composition: Active ingredient - pendimethalin 33% W/V

Formula: $C_{13}H_{19}N_3O_4$

Molecular weight: 281.312 g/mol

Structural formula:



4. FIRST AID MEASURES

Description of first aid measures

If poisoning occurs, contact a Doctor.

In case of swallowed:

Never induce vomiting or give anything by mouth if the victim is unconscious or having convulsions. Do not induce vomiting. Have person sip a glass of water if able to swallow.

In case of eyes contact:

Hold the eyes open and rinse slowly and gently with water for 15 to 20 minutes, remove contact lenses, if present, after first 5 minutes, then continue.

In case of skin contact:

Rinse skin immediately with plenty of water and soap for 15 to 20 minutes.

In case of Inhaled contact:

If vapors or mists have been inhaled and irritation has developed, remove the affected individual into fresh air and keep the person calm, observe until recovered. If breathing has stopped, give artificial respiration. **ADVICE TO DOCTOR:** Treat symptomatically mainly for hydrocarbon solvent ingestion. Note the nature of this product. If product has been aspirated into the lungs (for example from vomiting), consider the possibility of chemical pneumonitis.



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5. FIRE FIGHTING MEASURES

Flash point: > 60 °C

Extinguishing Media:

Foam, CO₂, dry chemical and water fog.

Flammability limit:

The product is non-flammable. When temperature >60°C, it will be flammable.

Unusual Fire and Explosion hazards:

Fire decomposition products may form toxic and corrosive mixtures in confined areas.

Hazardous Decomposition Products:

May decompose at 200 °C.

Special Fire Fighting Procedures:

If a significant quantity of this product is involved in a fire, call the fire brigade. Immediately evacuate the area of unnecessary personnel. When fighting fires involving significant quantities of this product, wear safety boots, non-flammable overalls, gloves, hat, goggles and respirator. All skin areas should be covered.

Materials to avoid:

Strong oxidizing agents.

6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

Action to take for spills and disposals: In event of a major spill, prevent spillage from entering drains or water courses. Call Fire Brigade or emergency services. Wear full protective clothing including face mask, face shield and gauntlets. All skin areas should be protected. Absorb onto vermiculite or other suitable absorbent material. Sweep up and shovel or collect recoverable product into labeled containers for recycling or salvage. After spills, wash area, preventing run off from entering drains. If material enters drains, advise emergency services. Launder all contaminated clothing before.

7. HANDLING AND STORAGE

Good industrial practice in housekeeping and personal hygiene should be followed.

Precautions to be taken in Handling and storage:

Precautions to be taken in handling and storage: Keep out of reach of children. Do not breathe vapors. Avoid contact with skin and swallowed. If swallowed, do not induce vomiting; seek medical advice immediately. Keep away from food, drink and animal foodstuffs. Keep the product at above 0°C. Keep containers tightly closed and do not store with foods, seeds, fertilizers or other pesticides.



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8. EXPOSURE CONTROL/PERSONAL PROTECTION

Exposure standards

No exposure limits have been set for this product.

Engineering controls

Natural ventilation only required. When handling the product in confined spaces it is advisable to use some forced ventilation to avoid build-up of solvent vapors.

Personal Protection

When opening the container and preparing spray or using the prepared spray, wear cotton overalls buttoned to the neck and wrist, a washable hat, elbow length PVC gloves and face shield or goggles. If product or spray gets on skin, immediately wash area with soap and water. After use and before eating, drinking or smoking, wash hands, arm and face thoroughly with soap and water. After each day's use, wash gloves, face shield and goggles and contaminated clothing.

Eyes Protection:

Protective eyewear is suggested when using this product.

Skin Protection:

Impermeable protective gloves should be worn when this product is being used to prevent irritation.

Respiratory Protection:

Use a dust mask or respiratory protection on account of this product.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical stage:	Homogenous liquid. No visible suspended, substance and sediment
Color:	Dark brown
Acidity (as H ₂ SO ₄):	≤ 0.5%
Vapor pressure:	4.0 mPa at 25 °C
Solubility in water:	0.3 Mg/L at 25°C
Specific gravity:	Approx. 1 g/ml

10. STABILITY AND REACTIVITY

Stability:	Stable for 2 years for the date of production in the room temperature (around 20°C)
Hazards polymerization:	Will not occur.
Incompatible Ities:	Will not occur.



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11. TOXICOLOGICAL INFORMATION

Information on Acute toxicity for end product, Tetraconazole:

LD50 (Oral, rat): > 2,000 mg/kg

LD50 (Dermal, rabbit): > 2,000 mg/kg

LC50 (Inhalation, rat, 4h): > 4.69 mg/L air (max attainable conc.)

Skin irritation (rabbit): mild irritant to the rabbit skin.

Eye irritation (rabbit): mild irritant to the rabbit skin.

Skin sensitization: in guinea pigs,

OECD harmonized classification system (August 14, 2001) - Negative

Magnusson and Kingman Rating of Allergenicity - Weak Sensitizer, Grade 1

Carcinogenicity data

Genetic toxicity: No mutagenic effect was found in various tests with microorganisms and mammals.

Carcinogenicity: in long term studies in mice in which the substance was given by feed, a carcinogenic effect was not observed. In long term studies in rats the substance induced thyroid tumors. A marked decrease in body weight gain and an increase in benign thyroid proliferate lesions were observed in the lifetime rat study at the highest dose tested. The substance is not considered to pose a carcinogenic risk at low human exposure levels.

Reproductive toxicity: In a three-generation reproductive study of rats tested at levels up to 250 mg/kg/day, there were slightly fewer offspring and they showed decreased weight gain from weaning to maturity. No effects were observed at 30 mg/kg/day. The results of animal studies gave no indication of a fertility impairing effect.

Teratogenic effects: No birth defects and no toxic effects on fetuses occurred when pregnant rats were given 500 mg/kg/day, the highest dose tested. No fetotoxic or teratogenic effects were seen at the highest dose tested (60 mg/kg/day) in a teratology study with rabbits, although maternal toxicity was seen at 30 mg/kg/day. No indication of a developmental toxic /teratogenic effect were seen in animal studies.

Mutagenic effects: Several mutagen city studies, including tests on live animals and mammalian and bacterial cell cultures, have all indicated that pendimethalin has no mutagenic activity.

12. ECOLOGICAL INFORMATION

Toxicity

Adopt good working practices, so that the product is not released into the environment.

Data referred to the mixture, pendimethalin:



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Fish – Acute toxicity:

Rainbow trout,

LC50 (96 h): 138 µg/L

Bluegill sunfish,

LC50 (96 h): 199 µg/L

Channel catfish,

LC50 (96 h): 420 µg/L

Invertebrates – Acute toxicity:

Water flea (Daphnia magna),

EC50 (48 h, static): 280 µg/L

The bioconcentration factor for this compound in whole fish is 5100, indicating a moderate potential to accumulate in aquatic organisms.

Birds – Toxicity: slightly toxic to birds

Bobwhite quail,

LC50 (8 days): > 3,149 mg/kg

Mallard ducks,

LC50 (5 days): 1,421 mg/kg

LC50 (8 days): > 10,388 mg/kg

Arthropod – Toxicity:

Bee,

LD50 (topical, 48 h): > 50 µg/bee

Other organism – Toxicity:

Crustaceans: northern pink shrimp, moderately toxic

13. DISPOSAL CONSIDERATIONS

Disposal method of empty, used containers by:

Do not contaminate waterways by cleaning of equipment or by disposal of wastes.

Untreated effluent should not be discharged where it will drain into lakes, streams, rivers or ponds.

Used containers disposal by:

Empty container retains product residue. Observe all hazard precautions. Do not distribute or make available, furnish or reuse empty container except for storage and shipment of original product. Remove all product residues from container by triple rinsing and puncture or otherwise destroy empty containers before disposal.



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14. TRANSPORT INFORMATION

The data provided in this section is for information only. Please apply the appropriate regulations to properly classify your shipment for transportation.

IMO INFORMATION:

Non-Bulk packaging: Environmentally hazardous substance, liquid, 9, UN 3082, PGIII

Bulk packaging: Environmentally hazardous substance, liquid, 9, UN 3082, PG III

UN No: 3082

Dangerous Goods Class: 9

Packaging Group: III

15. REGULATORY INFORMATION

NOTICE: The information herein is presented in good faith and believed to be accurate as of the effective date shown above. However, no warranty, express or implied, is given. Regulatory requirements are subject to change and may differ from one location to another; it is the buyer's responsibility to ensure that its activities comply with national, or provincial, and local laws.

This product contains the following substance(s) listed as "Hazardous Substances", which may require reporting of releases:

Chemical Name	CAS Number	Content of Active ingredient
Pendimethalin	40487-42-1	330 g/L

16. OTHER INFORMATION

The information herein is given in good faith, but no warrant, express or implied, is made.